9

Sampling

9.1 Introduction

Sampling is a process of converting a continuous-time signal into discrete-time signal. After sampling, the signal is defined at discrete instants of time and the time interval between two subsequent sampling instants is called sampling interval. In the process of sampling, one of the important factors that we have to consider is the signal sampling rate must be kept sufficiently high so that the original signal can be reconstructed from its samples.

9.2 Analog to Digital Conversion

Most of the signals encountered in nature are continuous-in-time. But in digital signal processing the signals are sampled and quantized at discrete-time instants and are represented as a sequence of 1 s and 0 s. For example, signals such as speech signal, ECG and EEG are electrical signals and therefore necessary to perform a conversion to digital representation. This can be done by an analog to digital converter. Fig.9.1 shows the ADC unit and its constituent parts. The sampler, samples the input signal with a sampling interval T, producing an output $x(nT)$. The signal $x(nT)$ is discrete-in-time but continuous in amplitude. The output of the sampler is applied to a quantizer. It converts $x(nT)$ into discrete-time, discrete-amplitude signal. After sampling and quantization, the final step in converting an analog signal to a form acceptable to digital computer is called coding. The coder maps each quantized sample value in digital word.

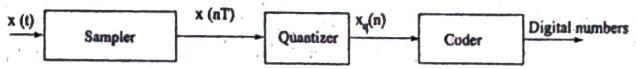


Fig. 9.1 Analog to digital conversion process

9.3 Sampling and Aliasing

In practice, sampling is performed by applying a continuous-time signal to an ADC whose output is a series of digital values. The primary concern in sampling

Multiple Choice Questions

1. Match the following and choose the correct combination

Group1	Group 2
E- Continuous and aperiodic signal	1. Fourier representation is continuous and aperiodic
F - Continuous and periodic signal	2. Fourier representation is discrete and aperiodic
G - Discrete and aperiodic signal	3. Fourier representation is continuous and aperiodic
H - Discrete and periodic signal	4. Fourier representation is discrete and periodic

(a) $E-3, F-2, G-4, H-1$ (b) $E-1, F-3, G-2, H-4$

(c) $E-1, F-2, G-3, H-4$ (d) $E-2, F-1, G-4, H-3$

2. A signal $x(n)=\sin(\Omega_0 n+\phi)$ is the input to a linear time invariant system having a frequency response $H(e^{j\Omega})$. If the output of the system A $x(n-n_0)$, then the most general form of $\angle H(e^{j\Omega})$ will be

(a) $-n_0\Omega_0+\beta$ for any arbitrary real β

(b) $-n_0\Omega_0+2\pi k$ for any arbitrary integer k

(c) $n_0\Omega_0+2\pi k$ for any arbitrary integer k

(d) $-n_0\Omega_0$

3. The direct evaluation DFT requires __ complex multiplications

(a) $N(N-1)$ (b) N^2 (c) $N(N+1)$ (d) $\dfrac{N(N-1)}{2}$

4. The DFT of the sequence $x(n)=\delta(n-n_0)$ is

(a) 1 (b) $e^{j2\pi kn_0}$ (c) $e^{-j2\pi n_0 k/N}$ (d) $e^{j2\pi n_0 k/N}$

5. The discrete Fourier transform of $x^*(n)$ is

(a) $X^*(k)$ (b) $X^*(-k)$ (c) $X(N-k)$ (d) $X^*(N-k)$

6. According to Parseval's Theorem $\sum_{n=0}^{N-1}|x(n)|^2=$

(a) $\sum_{k=0}^{N-1}|X(k)|^2$ (b) $\sum_{k=-\infty}^{\infty}|X(k)|^2$ (c) $\frac{1}{N}\sum_{k=0}^{N-1}|X(k)|^2$ (d) $|X(k)|^2$

7. The number of complex multiplications required to calculate N-point DFT using radix-2 DIT-FFT algorithm is

(a) $N\log_2 N$ (b) $\dfrac{N}{2}\log_{10}^{N}$ (c) $N\log_{10}N$ (d) $\dfrac{N}{2}\log_2 N$

8. If $X(k)$ is DFT of a sequence $x(n)$ then DFT of real part of $x(n)$ is

(a) $X^*(k)$ (b) $X^*(N-k)$

(c) $\dfrac{1}{2}[X(k)+X^*(N-k)]$ (d) $\dfrac{1}{2}[X(k)-X^*(N-k)]$

Answers to Multiple Choice Questions

1. (c) $E-1, F-2, G-3, H-4$
2. (b)
3. (b) N^2
4. (c) $e^{-j2\pi n_0 k/N}$
5. (d) $X^*(N-k)$
6. (c) $\frac{1}{N}\sum_{k=0}^{N-1}|X(k)|^2$
7. (d) $\frac{N}{2}\log_2 N$
8. (c) $\frac{1}{2}[X(k)+X^*(N-k)]$

is to select the sampling rate (number of samples per second) that preserve the information contained in the continuous-time signal.

If T is the time interval between successive samples, then the discrete-time signal can be represented by the relation

$$x(nT) = x(n) \quad -\infty < n < \infty \tag{9.1}$$

The time interval between successive samples is called the sampling period and its reciprocal $\frac{1}{T} = f_s$ is called sampling rate.

Let us consider a sinusoidal signal

$$x(t) = \sin \Omega t \tag{9.2}$$

The sampled sinusoidal sequence can be obtained by substituting $t = nT$ in Eq. (9.2). That is,

$$\begin{aligned} x(nT) &= \sin \Omega nT \\ &= \sin \omega n \end{aligned} \tag{9.3}$$

where

$$\omega = \Omega T \tag{9.4}$$

The fundamental difference between the continuous-time signal and discrete-time signal is the range of frequency variables Ω and ω. The range of frequency variable Ω in a continuous-time signal is given by $-\infty < \Omega < \infty$, whereas the range of frequency variable ω in a discrete-time signal is given by $-\pi < \omega < \pi$. That is, the periodic sampling of a continuous-time signal maps the infinite frequency variable Ω into a finite frequency variable ω.

If we imagine that the sinusoid of Eq. (9.2) is the input of an ADC, then the output of the A/D converter is given by Eq. (9.3), where the sampling frequency of the ADC is $\Omega_s = \frac{2\pi}{T}$. Let us study the output of ADC as a function of Ω. The output of ADC at frequency that differs by multiple of $\frac{2\pi}{T}$ is given by

$$\begin{aligned} x(n) &= \sin\left[(\Omega + \frac{2\pi M}{T})nT\right] \qquad M = 1,2,\ldots \\ &= \sin[(\Omega nT + 2\pi Mn)] = \sin \Omega nT \end{aligned} \tag{9.5}$$

which is identical to Eq. (9.3)

That is, the signals $\sin \Omega t$ and $\sin(\Omega + \Omega_s)t$ produce the same output samples. It means that a sequence $x(n)$ obtained by sampling the sinusoidal signal of Ω rad/sec (or f Hz), also exactly represents sine waves at other frequencies namely $\left(\Omega + \frac{2\pi k}{T}\right)$ or $(f + kf_s)$. That is, when the signal is sampled at a rate of f_s sample/sec, and if k is a positive or negative integer, we cannot distinguish between the sampled values of a sine wave of frequencies f Hz and $(f + kf_s)$ Hz. Thus an infinite number of continuous-time sinusoids are represented by the same set of samples. As a result an ambiguity exists as to which continuous-time signal $x(t)$ these values represent. To illustrate the effect of Eq. (9.5) let us consider two sinusoidal signals of frequencies 10 Hz and 40 Hz, which are applied to an ADC with a sampling frequency of 30 Hz. The corresponding discrete-time signals are

$$x_1(nT) = \sin \frac{2\pi(10)n}{30} = \sin 2\pi \left(\frac{n}{3}\right) \tag{9.6}$$

$$\begin{aligned} x_2(nT) &= \sin \frac{2\pi(40)n}{30} = \sin 2\pi \left(\frac{4n}{3}\right) \\ &= \sin 2\pi \left[1 + \frac{1}{3}\right] n = \sin \left[2\pi + \frac{2\pi}{3}\right] n \\ &= \sin \frac{2\pi}{3} n \end{aligned} \tag{9.7}$$

x	0	1	2	3	4	5
$x_1(nT) = \sin\left(\frac{2\pi n}{3}\right)$	0	0.866	−0.866	0	0.866	−0.866
$x_2(nT) = \sin\left(\frac{3\pi n}{3}\right)$	0	0.866	−0.866	0	0.866	−0.866

Since $x_1(nT) = x_2(nT)$ there is some ambiguity as to whether these sample values correspond to 10 Hz signal or 40 Hz signal. Similarly if the signal frequency is 20 Hz, then

$$\begin{aligned} x_3(nT) &= \sin\left[2\pi \left(\frac{20}{30}\right) n\right] = \sin 2\pi \left[1 - \frac{1}{3}\right] n \\ &= -\sin\left[\frac{2\pi}{3}\right] n \end{aligned} \tag{9.8}$$

with the same frequency as $x_1(nT)$. Since $x_3(nT)$ and $x_2(nT)$ yield exactly the same values as $x_1(nT)$ when the three signals are sampled at 30 samples per second, we can say that the frequencies 40 Hz and 20 Hz are alias of the frequency 10 Hz. Similarly all the sinusoids with frequencies $(f + 30k)$ $k = \pm 1, \pm 2, \pm 3, \ldots$ sampled at 30 samples per second are all aliases of 10 Hz. That is, these high frequency components can be mistaken for the lower frequency

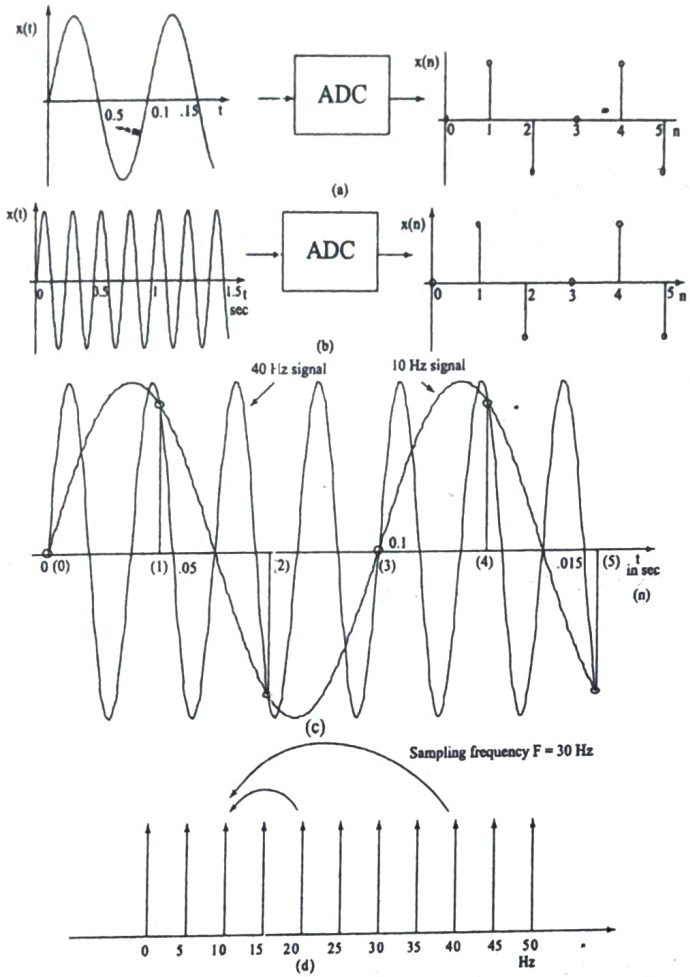


Fig. 9.2 (a) and (b) The output of ADC producing samples for different input waveforms (c) Two different sinusoidal signals having same sample values (d) Illustration of aliasing

components. This concept is illustrated in Fig.9.2, from which we can observe that all the input frequencies above 15 Hz will appear as the frequencies below 15 Hz.

In general if the sampling frequency is Ω_s (or f_s), all signals with frequencies above $\frac{\Omega_s}{2}$ (or $\frac{f_s}{2}$) cause aliasing. This aliasing can be avoided if the input signal frequencies are below one half of the sampling frequency. This frequency $\frac{\Omega_s}{2}$ (or $\frac{f_s}{2}$) is called the Nyquist frequency Ω_N (or f_N). Another name for $\frac{\Omega_s}{2}$ is the folding frequency, because it folds higher frequencies back down in the range $-\frac{\Omega_s}{2}$ to $\frac{\Omega_s}{2}$.

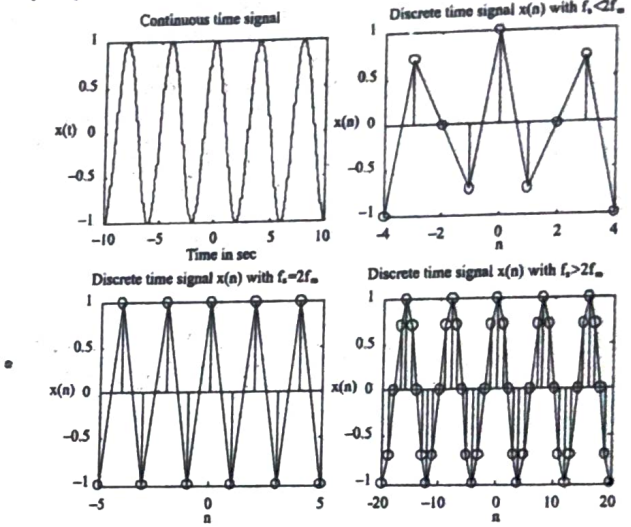


Fig. 9.3 (a) A continuous-time signal (b) Samples of continuous-time signal shown with sampling rate less than Nyquist rate where the original signal cannot be constructed from the samples (c) & (d) Samples of continuous-time signal with sampling rate equal and greater than Nyquist rate respectively where the original CT signal can be reconstructed.

If we restrict our spectral band of interest to the frequency range of $\pm\frac{\Omega_s}{2}$ and sample the input signal with a sampling rate Ω_s then the aliasing can be avoided. If the highest frequency present in the input signal is f_m, then the sampling rate f_s must be greater than or equal to $2f_m$. That is,

$$f_s \geq 2f_m \tag{9.9}$$

Thus if the maximum frequency component present in an analog signal is 3 KHz, then to preserve all the information in the signal, it should be sampled at 6 KHz or more.

To demonstrate the sampling theorem consider the sinusoidal signal shown in Fig.9.3(a) which is sampled at a rate less than Nyquist rate as shown in Fig.9.3(b). From Fig.9.3(b) we can find that it is not possible to reconstruct uniquely the original sinusoid because there exists several waves passing through the sampled values. Therefore sampling with a rate less than Nyquist rate renders useless sampled values. Fig.9.3(c) shows the samples obtained by sampling the continuous time signal at a rate equal to Nyquist rate from which we can reconstruct the original sinusoid. Similarly by using the samples shown in Fig.9.3(d) (Sampling rate greater than Nyquist rate), we can reconstruct the original sinusoidal signal.

9.4 Impulse sampling

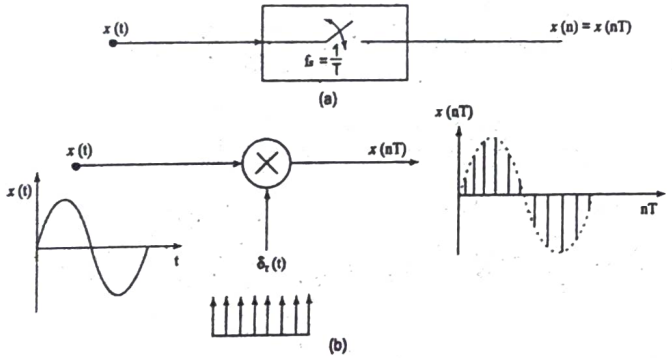


Fig. 9.4 Sampling operation.

The sampling operation can be represented by a fictitious switch shown in Fig.9.4. The switch is closed for a very short interval of time τ, during which the signal is available at the output. Therefore if the input is $x(t)$, the output is $x(nT), n = 0, \pm 1, \pm 2 \ldots$ and $x(nT)$ is called the sample sequence of $x(t)$ where T is the time interval between successive samples and the sampling frequency is given by $f_s = \frac{1}{T}$ Hz. The discrete-time signal $x(nT)$ obtained by sampling is shown in Fig. 9.4(b).

The another way of representing the discrete-time signal is

$$x_s(t) = x(t)\delta_T(t) \tag{9.10}$$

where

$$\delta_T(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT) \tag{9.11}$$

That is the function $x_s(t)$ is the product of $x(t)$ and the train of impulses. Therefore

$$x_s(t) = \sum_{n=-\infty}^{\infty} x(nT)\, \delta(t - nT) \tag{9.12}$$

which is a sequence of impulses at $t = nT$ with weight $x(nT)$. Thus $x_s(t)$ can be considered as a continuous-time representation of the discrete-time sequence $x(nT)$. Applying Fourier transform to Eq. (9.12) yields

$$F[x_s(t)] = F\left[\sum_{n=-\infty}^{\infty} x(nT)\, \delta(t - nT)\right]$$

$$= \sum_{n=-\infty}^{\infty} x(nT) F[\delta(t - nT)] \tag{9.13}$$

We have

$$F[\delta(t - nT)] = \int_{-\infty}^{\infty} \delta(t - nT) e^{-j\omega t} dt = e^{-j\omega nT}. \tag{9.14}$$

Thus,

$$X(e^{j\omega}) = F[x_s(t)] = \sum_{n=-\infty}^{\infty} x(nT) e^{-j\omega nT} \tag{9.15}$$

From Eq. (6.120) we have

$$\delta_T(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT) = \frac{1}{T} \sum_{m=-\infty}^{\infty} e^{jm\Omega_0 t} \tag{9.16}$$

where $\Omega_0 = \dfrac{2\pi}{T}$.

Substituting Eq. (9.16) in Eq. (9.10) we obtain

$$x_s(t) = x(t)\left[\frac{1}{T} \sum_{m=-\infty}^{\infty} e^{jm\Omega_0 t}\right] = \frac{1}{T} \sum_{m=-\infty}^{\infty} x(t) e^{jm\Omega_0 t}. \tag{9.17}$$

Now

$$F[x_s(t)] = \frac{1}{T} \sum_{m=-\infty}^{\infty} F[x(t) e^{jm\Omega_0 t}] \tag{9.18}$$

If $F[x(t)] = X(j\Omega)$ then

$$F[x(t) e^{jm\Omega_0 t}] = X[j(\Omega - m\Omega_0)] \tag{9.19}$$

Substituting Eq. (9.19) in Eq. (9.18), we get

$$F[x_s(t)] = \frac{1}{T}\sum_{m=-\infty}^{\infty} X[j(\Omega - m\Omega_0)] \tag{9.20}$$

$$= \frac{1}{T}\sum_{m=-\infty}^{\infty} X\left[j\left(\Omega - \frac{2\pi m}{T}\right)\right] \tag{9.21}$$

Thus the Fourier transform of the sampled signal is given by an infinite sum of shifted replicas of the Fourier transform of the original signal.

Now consider a signal $x(t)$ band limited to f_m. That is the highest frequency component of $x(t)$ is f_m. Then

$$X(j\Omega) = 0 \quad \text{for} \quad |\Omega| > \Omega_m \qquad (\Omega_m = 2\pi f_m) \tag{9.22}$$

In Eq. (9.22), the term $X\left[j\left(\Omega - \frac{2\pi m}{T}\right)\right]$ is the shifting of $X(j\Omega)$ from $\Omega = 0$ to $\Omega = \frac{2\pi m}{T}$. Hence $X_s(j\Omega)$ is the sum of shifted replicas of $\frac{X(j\Omega)}{T}$ centering at $\frac{2\pi m}{T}, m = 0, \pm 1, \pm 2 \ldots$. The Fig. 9.5 shows the plot of $\frac{X(j\Omega)}{T}$ for different values of $\frac{\pi}{T}$.

We see that if $\frac{\pi}{T} > \Omega_m$, the replicas will not overlap as in Fig. 9.5(c) and as a result, the frequency spectrum of $TX_s(j\Omega)$ in frequency range $\left(-\frac{\pi}{T}, \frac{\pi}{T}\right)$ is identical to $X(j\Omega)$. From Fig.9.5(c) we observed that there is a gap between successive replicas. This gap is between the maximum frequency component Ω_m and the lower limit of the replica centred at Ω_s. The frequency range from Ω_m to $\frac{2\pi}{T} - \Omega_m$ is called a guard band and is always required in practice. The presence of guard band avoids the need for a filter that has sharp cut off characteristics. Now the frequency spectrum $X(j\Omega)$ can be recovered from $X_s(j\Omega)$ by using a lowpass filter with cut-off frequency at $\Omega_c = \frac{\pi}{T}$. The same explanation holds for $\frac{\pi}{T} = \Omega_m$. If $\frac{\pi}{T} < \Omega_m$, the successive frequency replicas will overlap as shown in Fig. 9.5(d) and as a result, the frequency spectrum $X(j\Omega)$ will not be recovered from the frequency spectrum of $TX_s(j\Omega)$. For example, consider an arbitrary point at Ω_1 which falls in the region of the overlap. The frequency spectrum at $\Omega = \Omega_1$ is the sum of two components, one component is $X_s(j\Omega_1)$ and the other component is $X_s\left[j\left(\frac{2\pi}{T} - \Omega_1\right)\right]$. This high frequency component $\left(\frac{2\pi}{T} - \Omega_1\right)$ is folded in about the folding frequency $\frac{\pi}{T}$ and appears as a low frequency at Ω_1. The super imposition of high frequency component on the low frequency is known as frequency aliasing. Because of aliasing, the spectrum $X(j\Omega)$ is no longer recoverable from the spectrum of $X_s(j\Omega)$.

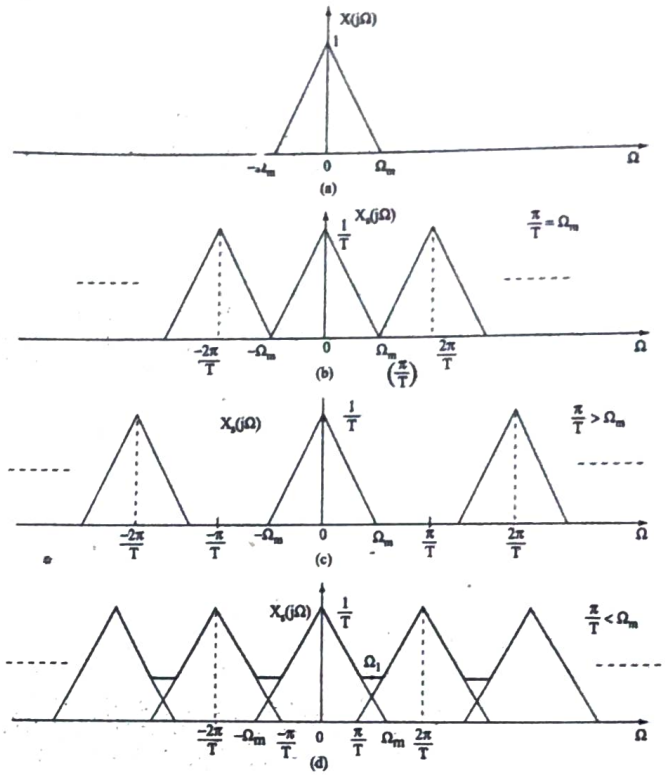


Fig. 9.5 (a) Frequency spectrum of continuous-time signal $x(t)$ (b) Frequency spectrum of $x(nT)$ with $\frac{\pi}{T} = \Omega_m$ (c) Frequency spectrum of $x(nT)$ with $\frac{\pi}{T} > \Omega_m$ (d) Frequency spectrum of $x(nT)$ with $\frac{\pi}{T} < \Omega_m$.

From Fig. 9.5 we can find that aliasing error can be prevented if the highest frequency component Ω_m in the signal $x(t)$ is less than or equal to $\frac{\pi}{T}$, that is, $\frac{\pi}{T} \geq \Omega_m$.

If $f_s = \frac{1}{T}$ then

$$\pi f_s \geq \Omega_m$$
$$\pi f_s \geq 2\pi f_m$$
or
$$f_s \geq 2f_m \quad (9.23)$$

That is, to avoid aliasing the sampling fequency must be greater than twice the highest fequency present in the signal.

9.5 Sampling Theorem

A band limited signal $x(t)$ with $X(j\Omega) = 0$ for $|\Omega| > \Omega_m$ is uniquely determined from its samples $x(nT)$, if the sampling fequency $f_s \geq 2f_m$, i.e., sampling frequency must be at least twice the highest frequency present in the signal.

Nyquist rate

The minimum rate at which a signal can be sampled and still be reconstructed from its samples is called the Nyquist rate. It is always equal to $2f_m$ where f_m is the maximum frequency component present in the signal.

A signal sampled at greater than Nyquist rate is said to be over-sampled and a signal sampled at less than its Nyquist rate is said to be under-sampled .

9.6 Anti Aliasing Filter

Sampling theorem states that a signal can be perfectly reconstructed if it is band limited. In practice, communication signals have frequency spectra consisting of low frequency components as well as high frequency noise components. When a signal is sampled, with sampling frequency f_s, all signals with frequency range higher than $\frac{\Omega_s}{2}$ appear as signal frequencies between 0 and $\frac{\Omega_s}{2}$ creating aliasing. Therefore, to avoid aliasing errors caused by the undesired high frequency signals, an analog lowpass filter, called an antialiasing filter is used prior to sampler as shown in Fig. 9.6.

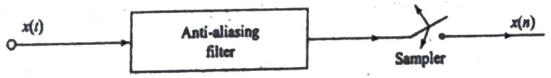


Fig. 9.6 Anti-aliasing filter prior to sampler.

Definition: A filter that is used to reject high frequency signals before it is sampled to reduce the aliasing is called an anti aliasing filter.

9.7 Pulse Sampling

In previous section we consider the sampling process in which the signal $x(t)$ is multiplied by a series train of impulses. However, such instantaneous sampling is hardly feasible. Instead a series of pulse having width τ, with a time period of T can be generated by changing the switch position between two points 1 and 2 as shown in Fig.9.7.

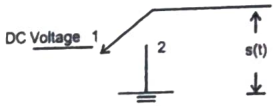


Fig. 9.7

Here the switch s is connected to position 1 for a time τ and it is grounded for a period of T. The resultant signal is shown in Fig.9.8.

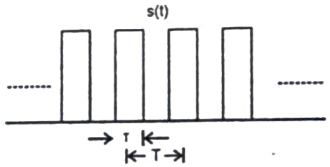


Fig. 9.8

If we replace the dc voltage by a signal $x(t)$ shown in Fig.9.9(a) then the output $x_s(t)$ follows the input signal for a time τ and this repeats for every T as shown in Fig.9.9(b).

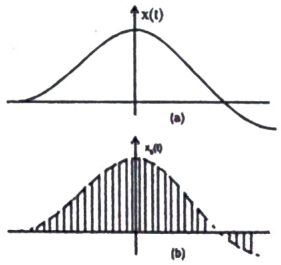


Fig. 9.9

The output $x_s(t)$ can be obtained by multiplying the signal $x(t)$ with $s(t)$. That is

$$x_s(t) = x(t)s(t) \tag{9.24}$$

Since $s(t)$ is a periodic signal it can be expressed in Fourier series as

$$s(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\Omega t} \tag{9.25}$$

$$\text{where } c_n = \frac{1}{T}\int_{-T/2}^{T/2} s(t)e^{-jn\Omega t}dt$$

The signal $s(t)$ can be expressed as sum of series of pulse having width τ. That is

$$s(t) = \sum_{k=-\infty}^{\infty} p(t-kT) \tag{9.26}$$

$$\text{where } p(t) = 1 \text{ for } |t| < \tau/2 \tag{9.27}$$
$$= 0 \text{ otherwise}$$

We have

$$c_n = \frac{1}{T}\int_{-T/2}^{T/2} s(t)e^{-jn\Omega t}dt$$

$$= \frac{1}{T}\int_{-\infty}^{\infty} p(t)e^{-jn\Omega t}\,dt$$

$$= \frac{1}{T}P(n\Omega_s) \tag{9.28}$$

$$\text{where } P(n\Omega_s) = F[p(t)]|_{\Omega=n\Omega_s}$$

$$s(t) = \frac{1}{T}\sum_{n=-\infty}^{\infty} P(n\Omega_s)e^{jn\Omega t} \tag{9.29}$$

and

$$X_s(j\Omega) = F[x_s(t)] = F\left[\frac{1}{T}\sum_{n=-\infty}^{\infty} P(n\Omega_s)x(t)e^{jn\Omega t}\right]$$

$$= \frac{1}{T}\sum_{n=-\infty}^{\infty} P(n\Omega_s)[X(j\Omega) * \delta(\Omega - n\Omega_s)]$$

$$\text{since } F[e^{jn\Omega t}] = \delta(\Omega - n\Omega_s)$$

$$X_s(j\Omega) = \frac{1}{T}\sum_{n=-\infty}^{\infty} P(n\Omega_s)X[j(\Omega - n\Omega_s)] \tag{9.30}$$

The pulse given in Fig.9.8 can be expressed as

$$\Pi\left(\frac{t}{\tau}\right) = 1 \text{ for } |t| \le \tau/2 \tag{9.31}$$
$$= 0 \text{ otherwise}$$

The Fourier transform of rectangular pulse is

$$c_n = \frac{\tau}{T}\sin\frac{\Omega_s n\tau}{2} \tag{9.32}$$

Substituting Eq.(9.32) in Eq.(9.29) we get $s(t) = \sum_{n=-\infty}^{\infty} \frac{\tau}{T}\sin\left(\frac{\Omega_s n\tau}{2}\right)e^{jn\Omega t}$

$$x_s(t) = x(t)s(t) = \frac{\tau}{T}\sum_{n=1}^{\infty}\text{sinc}\left(\frac{\Omega_s n\tau}{2}\right)\cos\Omega_s t + \frac{\tau}{T}$$

If $x(t)$ has a spectrum as shown in Fig.9.5(a) then $X_s(j\Omega)$, has the spectrum as shown in Figure.9.10.

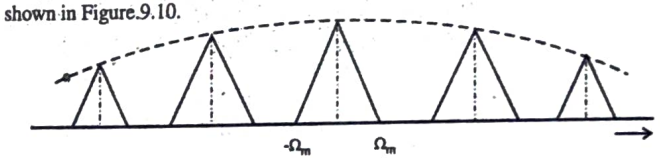


Fig. 9.10

From the frequency spectrum shown in Fig 9.10 we can find that the amplitude of the various harmonics are not the same when we use pulse sampling. It is clear that $X(j\Omega)$ can be recovered from $X_s(j\Omega)$ by using a lowpass filter.

9.8 Flat-Top sampling

In natural sampling discussed in section (9.7) the tops of the pulses follow the wave form of the signal. Such type of signals are not frequently employed. Instead flat-topped pulses (Fig.9.12) generated using a sample-and hold circuit are customarily used. The sampled-and-hold circuit consists of two switches S_1 and S_2 as shown in Fig.9.11. A capacitor is placed across the switch 1 and the capacitor hold the signal voltage until discharged by a pulse applied to G_2.

The sampled waveform is given by

$$x_p(t) = \sum_k x(kT)p(t-kT)$$

We know

$$p(t) * \delta(t-kT) = p(t-kT)$$

$$\Rightarrow x_p(t) = p(t) * \sum_k x(kT)\delta(t-kT)$$

$$= p(t) * x_\delta(t)$$

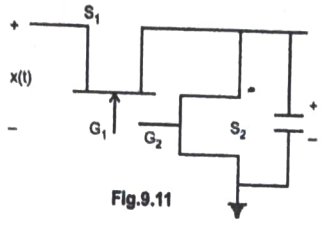


Fig.9.11

The Fourier transform of the signal is

$$X_p(j\Omega) = P(j\Omega)\left[\frac{1}{T}\sum_n X(j(\Omega - n\Omega_s))\right]$$

$$= P(j\Omega)X_\delta(j\Omega)$$

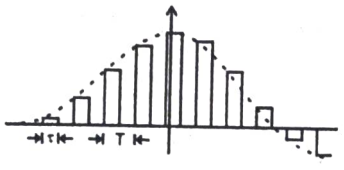


Fig. 9.12

From above equation we find that the flat-top sampling is equivalent to passing an ideal sampled wave through a circuit with transfer function $P(j\Omega)$. This circuit which is a lowpass filter attenuates the high frequency components of the signal spectra. This loss of high frequency component is called aperture effect. The aperture effect results from the fact that the spectrum is multiplied by $P(j\Omega)$, which is a sin$c$ function. The magnitude of sin$c$ function decreases slowly with increasing Ω. When this function is multiplied by $X(j\Omega)$, the high frequency components attenuate resulting distortion. The distortion reduces as τ decreases and approaches zero as $\tau \to 0$ (impulse sampling).

One way to reduce the apperture effect distortion is to connect an equalizer in cascade with the output of lowpass filter. An equalizer is a passive network with transform function of the form $\frac{x}{\sin x}$, a form inverse to the form of lowpass filter.

Review Questions

1. Distinguish between instantaneous sampling, natural sampling and flat-top sampling.
2. What is aperture effect? How it can be reduced?

9.9 Signal Reconstruction

In the preceding section, we have discussed that a band limited signal $x(t)$ can be reconstructed from its samples if the sampling rate is above Nyquist rate. Let the sampled signal $x_s(t)$ with sampling period T second is

$$x_s(t) = \sum_{n=-\infty}^{\infty} x(nT)\delta(t-nT) \tag{9.33}$$

The Fourier transform of $x_s(t)$ is given by

$$X(e^{j\alpha}) = \frac{1}{T}\sum_{m=-\infty}^{\infty} X\left[j\left(\Omega - \frac{2\pi m}{T}\right)\right]. \tag{9.34}$$

The frequency spectrum $X(e^{j\omega})$ consists of an infinite sum of shifted replicas of the frequency spectrum of the original signal.

To get the original spectrum the signal $x_s(t)$ is to be passed through a reconstruction (low-pass) filter with impulse response $h(t)$.

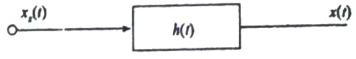


Fig. 9.13

The original signal can be obtained by convolving the sampled signal $x_s(t)$ and the impulse response $h(t)$ of the filter. That is,

$$x(t) = x_s(t) * h(t) = \left[\sum_{n=-\infty}^{\infty} x(nT)\delta(t-nT)\right] * h(t)$$

$$= \int_{-\infty}^{\infty}\sum_{n=-\infty}^{\infty} x(nT)\,\delta(\lambda - nT)\,h(t-\lambda)d\lambda$$

$$= \sum_{n=-\infty}^{\infty} x(nT)\int_{-\infty}^{\infty}\delta(\lambda - nT)\,h(t-\lambda)\,d\lambda$$

$$= \sum_{n=-\infty}^{\infty} x(nT)\,h(t-nT) \tag{9.35}$$

$\therefore \quad \delta(\lambda - nT) = 1 \text{ for } \lambda = nT$
$\qquad\qquad = 0 \text{ for } \lambda \neq nT$

The frequency response of the lowpass filter can be expressed by

$$H(j\Omega) = T \quad \text{for} \quad |\Omega| \leq \frac{\Omega_s}{2}$$

$$= 0 \quad \text{for} \quad |\Omega| > \frac{\Omega_s}{2} \tag{9.36}$$

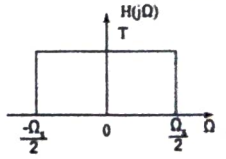


Fig.9.14 Frequency response of a lowpass filter .

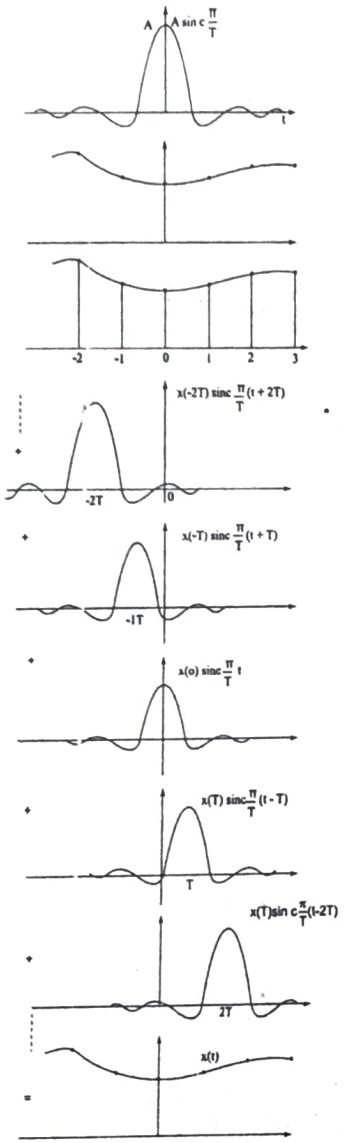


Fig. 9.15 Reconstruction of band limited signals from its samples.

The impulse response of the lowpass filter is

$$h(t) = \frac{1}{2\pi}\int_{-\frac{\Omega_s}{2}}^{\frac{\Omega_s}{2}} H(j\Omega)e^{j\Omega t}d\Omega$$

$$= \frac{T}{2\pi}\int_{-\frac{\Omega_s}{2}}^{\frac{\Omega_s}{2}} e^{j\Omega t}d\Omega = \frac{T}{\pi t}\sin\frac{\Omega_s t}{2}$$

$$= \frac{T}{\pi t}\sin\frac{\pi t}{T}$$

$$= \frac{T}{\pi}\frac{\sin\frac{\pi t}{T}}{t} \tag{9.37}$$

$$h(t-nT) = \frac{T}{\pi}\frac{\sin\frac{\pi}{T}(t-nT)}{(t-nT)} \tag{9.38}$$

Substituting Eq. (9.38) in Eq. (9.35) yields

$$x(t) = \sum_{n=-\infty}^{\infty} x(nT)\,\mathrm{sinc}\frac{\pi}{T}(t-nT) \tag{9.39}$$

The response $x(t)$ given by the formula is not suited for practical implementation. Notice that the sinc function spreads over entire time axis. To reconstruct the signal at time $t = t_0$, we must know all the sample values including those for $nT > t_0$. That is $x(t_0)$ requires the knowledge of future values of $x(nT)$. So it represents a non-causal system. Hence this type of reconstruction is not suited for real time applications. Therefore, we need a different approach to convert digital signal to analog signal. One such approach is shown in Fig. 9.16. Here the digital to analog converter produces an output voltage that is proportional to the value of the binary word. The output of DAC is applied to an interpolator which converts the output samples of DAC into an analog signal. There are several interpolators.

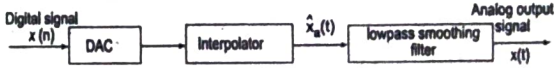


Fig. 9.16 Digital to Analog conversion.

Zero order hold

One of the most widely used interpolator is zero order hold. In this technique a given sample is held for an interval until the next sample is received. Mathematically

$$\hat{x}_a(t) = x(n) \quad \text{for} \quad nT \le n \le (n+1)T \quad n = 0,1,2,\ldots. \tag{9.40}$$

In particular,

$$\begin{aligned} \hat{x}_a(t) &= x(0) \quad \text{for} \quad 0 \le t < T \\ &= x(T) \quad \text{for} \quad T \le t < 2T \\ &= x(2T) \quad \text{for} \quad 2T \le t < 3T. \end{aligned} \tag{9.41}$$

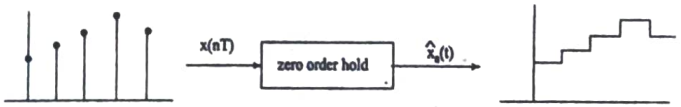


Fig. 9.17 Zero order hold.

The impulse response of zero order hold is given by

$$\begin{aligned} h(t) &= 1 \quad 0 \le t \le T \\ &= 0 \quad \text{otherwise} \end{aligned} \tag{9.42}$$

Transfer function of zero order hold

The output of zero order hold is expressed in terms of $h(t)$. That is the output $\hat{x}_a(t)$ is the convolution of $x(nT)$ and $h(t)$

$$\hat{x}_a(t) = \sum_{n=-\infty}^{\infty} x(nT)\, h(t-nT) \tag{9.43}$$

h(t) in Eq. (9.42) can be expressed as

$$h(t) = u(t) - u(t-T) \tag{9.44}$$

from which

$$h(t-nT) = u(t-nT) - u[t-(n+1)T] \tag{9.45}$$

Substituting Eq. (9.45) in Eq. (9.43) yields

$$\hat{x}_a(t) = \sum_{n=-\infty}^{\infty} x(nT)\left[u(t-nT) - u[t-(n+1)T]\right]$$

Taking Laplace transform on both sides

$$L[\hat{x}_a(t)] = \sum_{n=-\infty}^{\infty} x(nT) L\left\{u(t-nT) - u[t-(n+1)T]\right\}$$

$$= \sum_{n=-\infty}^{\infty} x(nT)\left[\frac{e^{-snT}}{s} - \frac{e^{-s(n+1)T}}{s}\right]$$

$$= \frac{1-e^{-sT}}{s} \sum_{n=0}^{\infty} x(nT) e^{-snT}$$

$$= \left[\frac{1-e^{-sT}}{s}\right] X^*(s)$$

$$\frac{X_a(s)}{X^*(s)} = \frac{1-e^{-sT}}{s}$$

The output of zero order hold consists of higher order harmonics. To remove these harmonics the output of zero order hold is applied to a lowpass filter as shown in Fig. 9.16. This filter is often called a smoothing filter because it tends to smooth the corners on the step approximations generated by the ZOH.

Review Questions

1. State sampling theorem. Also discuss aliasing by taking an example.
2. What is meant by aliasing? What is an anti-aliasing filter?
3. Let $x(t)$ be a real valued band-limited signal specified by

$$X(j\Omega) = 0 \qquad |\Omega| > \Omega_m$$

Show that $x(t)$ can be expressed as

$$x(t) = \sum_{k=-\infty}^{\infty} x(kT) \frac{\sin \Omega_m (t-kT)}{\Omega_m (t-kT)}$$

4. Let $x(t)$ be a real signal valued band limited signal whose Fourier transform can be represented as

$$X(j\Omega) = 0 \qquad |\Omega| > \Omega_m$$

Let the sample signal $x_s(t) = x(t)\delta_T(t)$ where $\delta_T(t) = \sum_{k=-\infty}^{\infty} \delta(t-kT_s)$

(a) Sketch $x(t)$ for $T < \frac{\pi}{\Omega_m}$ and for $T_s > \frac{\pi}{\Omega_m}$.

(b) Find and sketch the Fourier transform $X_s(j\Omega)$ of $x_s(t)$ for $T < \frac{\pi}{\Omega_m}$ and for $T > \frac{\pi}{\Omega_m}$.

(c) What is the effect of undersampling?

Solved Problem 9.1 A signal having a spectrum ranging from near *dc* to 10 KHz is to be sampled and converted to discrete form. What is the minimum number of samples per second that must be taken to ensure recovery?

Solution:

Given $f_m = 10\,\text{KHz}$

From Nyquist rate, the minimum number of samples per second that must be taken to ensure recovery is

$$f_s = 2f_m$$
$$= 20{,}000 \quad \text{samples/sec}$$

Solved Problem 9.2 If the Nyquist rate of a signal $x(t)$ is Ω_0, then what is the Nyquist rate of the following signals

(a) $x(t) - x(t-1)$

(b) $x(t)\cos\Omega_0 t$.

Solution:

(a) The Nyquist rate of $x(t)$ is Ω_0. Therefore the highest frequency component present in the signal is $\frac{\Omega_0}{2}$. That is $|X(j\Omega)| = 0$ for $\Omega > \frac{\Omega_0}{2}$. Let $y_1(t) = x(t) - x(t-1)$. The Fourier transform of $x(t) - x(t-1)$ is $X(j\Omega) - e^{-j\Omega}X(j\Omega)$

$$\Rightarrow Y_1(j\Omega) = X(j\Omega) - e^{-j\Omega}X(j\Omega)$$

We can find that $|Y_1(j\Omega)| = 0$ for $\Omega > \frac{\Omega_0}{2}$. Therefore Nyquist rate of $y_1(t)$ is Ω_0.

Let $y_2(t) = x(t)\cos\Omega_0 t$; $\quad Y_2(j\Omega) = \frac{1}{2}X[j(\Omega - \Omega_0)] + \frac{1}{2}X[j(\Omega + \Omega_0)]$

(b) The maximum frequency component in $x(t)$ is Ω_0. When the signal $x(t)$ is multiplied with $\cos\Omega_0 t$ the maximum frequency component present in the signal is $\left(\Omega_0 + \frac{\Omega_0}{2}\right) = 3\Omega_0/2$. Therefore the Nyquist rate is $3\Omega_0$.

Solved Problem 9.3 The signal $x(t) = 10\cos(10\pi t)$ is sampled at a rate 8 samples per second. Plot the amplitude spectrum for $|\Omega| \le 30\pi$. Can the original signal can be recovered from samples? Explain.

Solution:

Given $x(t) = 10\cos(10\pi t)$

$$F[x(t)] = 10\{\pi[\delta(\Omega + 10\pi) + \delta(\Omega - 10\pi)]\}$$

$$\boxed{F[\cos(\Omega_0 t] = \pi[\delta(\Omega + \Omega_0) + \delta(\Omega - \Omega_0)]}$$

The amplitude spectrum of $x(t)$ its shown in Fig. 9.18.

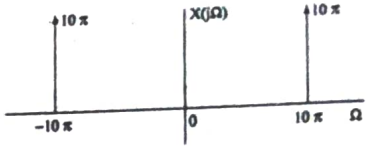


Fig. 9.18

The sampling rate $f_s = 8\,\text{Hz}$; $\quad \Omega_s = 2\pi f_s = 2\pi(8) = 16\pi$

$$\Omega_m = 2\pi f_m = 10\pi \Rightarrow f_m = 5\,\text{Hz}$$

$$\text{Nyquist rate} = 2f_m = 10\,\text{Hz}$$

The sampling rate is less than Nyquist rate. So, the original signal cannot be recovered from the samples.

The frequency spectra of sampled $x(t)$ is given by

$$X_s(j\Omega) = \frac{1}{T}\sum_{n=-\infty}^{\infty} 10\{\pi[\delta(\Omega + 10\pi - n\Omega_s] + \pi[\delta(\Omega - 10\pi - n\Omega_s)]\} \quad \left(\Omega_s = \frac{2\pi}{T}\right)$$

(use Eq. 9.1

$$= 80\pi\sum_{n=-\infty}^{\infty}[\delta(\Omega + 10\pi - 16n\pi) + \delta(\Omega - 10\pi - 16n\pi)]. \quad \left(\because \frac{1}{T} = 8\right)$$

The plot of amplitude spectrum for $|\Omega| \le 30\pi$ is shown in Fig. 9.19.

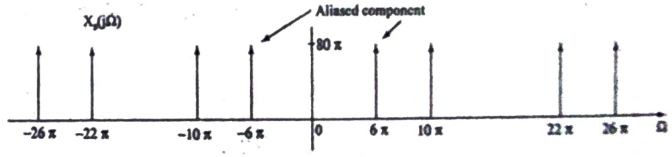


Fig. 9.19

Solved Problem 9.4 Two signals $x_1(t)$ and $x_2(t)$ are multiplied together and the product is sampled by a periodic impulse train $p(t) = \sum_{n=-\infty}^{\infty} \delta(t-nT)$. If the signal $x_1(t)$ and $x_2(t)$ are band limited to Ω_1 and Ω_2 respectively; That is

$$X_1(j\Omega) = 0 \text{ for } |\Omega| > \Omega_1$$
$$X_2(j\Omega) = 0 \text{ for } |\Omega| > \Omega_2$$

determine the maximum sampling interval T that recover the signal from its samples.

Solution:

Let the product of signals $x_1(t)$ and $x_2(t)$ is equal to $y(t)$. That is

$$y(t) = x_1(t)x_2(t)$$
$$Y(j\Omega) = \frac{1}{2\pi}[X_1(j\Omega) * X_2(j\Omega)]$$

Given $X_1(j\Omega) = 0$ for $|\Omega| > \Omega_1$ and $X_2(j\Omega) = 0$ for $|\Omega| > \Omega_2$. The maximum frequency component present in $Y(j\Omega)$ is $\Omega_1 + \Omega_2$. The Nyquist rate is $2(\Omega_1 + \Omega_2)$. The maximum sampling interval $T = \frac{1}{2f_m} = \frac{1}{2\frac{(\Omega_1+\Omega_2)}{2\pi}} = \frac{\pi}{(\Omega_1+\Omega_2)}$

Practice Problem 9.1

Two signals $x_1(t)$ and $x_2(t)$ are band limited to Ω_1 and Ω_2 respectively where

$$X_1(j\Omega) = 0 \text{ for } |\Omega| > 500\pi$$
$$X_2(j\Omega) = 0 \text{ for } |\Omega| > 1000\pi$$

The signal $y(t)$ is generated by convolving the signals $x_1(t)$ and $x_2(t)$, that is

$$y(t) = x_1(t) * x_2(t)$$

The signal $y(t)$ is sampled using Impulse train sampling. Find the range of values for sampling period T which ensures that $y(t)$ is recoverable from its samples.

Solved Problem 9.5 A signal $x(t) = \text{sinc}(150\pi t)$ is sampled at a rate of (a) 100 Hz (b) 200 Hz (c) 300 Hz. For each of these three cases, Explain if you can recover the signal $x(t)$ from the sampled signal.

Solution:

Given $x(t) = \text{sinc}(150\pi t)$

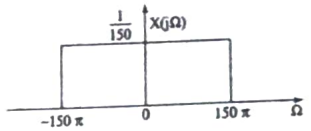


Fig. 9.20

The spectrum of the signal $x(t)$ is a rectangular pulse with a band width (maximum frequency component) of 150π rad/sec as shown in Fig. 9.20.

From Fig. 9.20 we have

$$2\pi f_m = 150\pi$$
$$f_m = 75 \text{ Hz}$$
$$\text{Nyquist rate } 2f_m = 150 \text{ Hz}.$$

a) In the first case the sampling rate is 100 Hz, which is less than Nyquist rate (under sampling). Therefore $x(t)$ cannot be recovered from its samples.

b) and (c)

In both cases the sampling rate is greater than Nyquist rate. Therefore $x(t)$ can be recovered from its samples.

Solved Problem 9.6 Draw $|X_s(j\Omega)|$ for the following cases if $x_s(t) = x(t)\delta_T(t)$ with sampling period T, where

$$\delta_T(t) = \sum_{n=-\infty}^{\infty} \delta(t-nT)$$

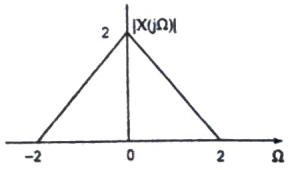


Fig. 9.21

(a) $T = \frac{\pi}{3}$ sec (b) $T = \frac{\pi}{2}$ sec (c) $T = 1$ sec

Solution:

(a) $T = \frac{\pi}{3}$ sec $\quad f_s = \frac{1}{T} = \frac{3}{\pi}$ Hz

$$\Omega_s = 2\pi f_s = 2\pi\left(\frac{3}{\pi}\right) = 6 \text{ rad/sec}$$

The spectra of $|X_s(j\Omega)|$ is shown in Fig. 9.22 which is periodic repetition of $X(j\Omega)$ for every 6 rad/sec.

From $X(j\Omega)$ we have

$$\Omega_m = 2 \text{ rad/sec}$$

$$f_m = \frac{1}{\pi} \text{Hz}$$

$$\text{Nyquist rate} = 2f_m = \frac{2}{\pi} \text{Hz}$$

$f_s > 2f_m$. Therefore the spectrum $X_s(j\Omega)$ is free from aliasing. Hence $X(i\Omega)$ can be recovered from $X_s(j\Omega)$ using an ideal low-pass filter of band width 2 rad/sec.

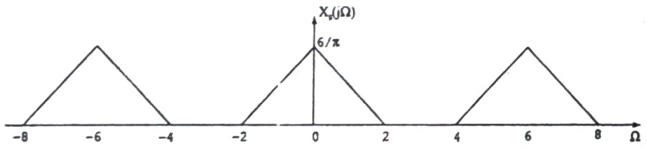


Fig. 9.22

(b) For this case the sampling period

$$T = \frac{\pi}{2} \text{ sec}$$

$$f_s = \frac{2}{\pi} \text{Hz}$$

$$\Omega_s = 2\pi f_s = 4 \text{ rad/sec}$$

Sampling frequency is equal to Nyquist rate. Hence $X(j\Omega)$ can be recovered from $X_s(j\Omega)$.

The spectra $|X_s(j\Omega)|$ is shown in Fig. 9.23. Here $X(j\Omega)$ repeats for every 4 rad/sec.

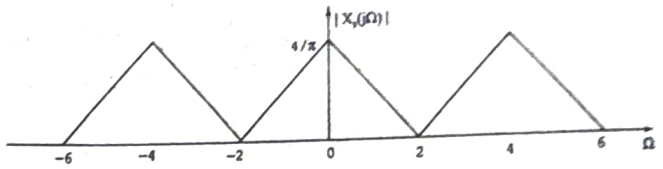


Fig. 9.23

(c) $T = \frac{2\pi}{3}$; $\quad f_s = \frac{3}{2\pi}$ sec $\quad \Omega_s = 3$ rad/sec.

The sampling frequency is less than the Nyquist rate. The spectra $X(j\Omega)$ repeats for every 3 rad/sec and the successive spectra overlap as shown in Fig. 9.24. Therefore $x(t)$ cannot be recovered from its samples.

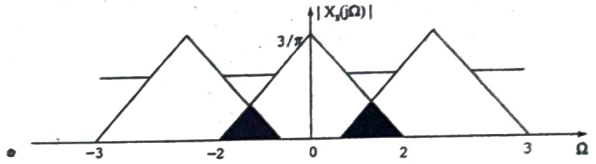


Fig. 9.24

Solved Problem 9.7 Consider the following sampling and reconstruction block

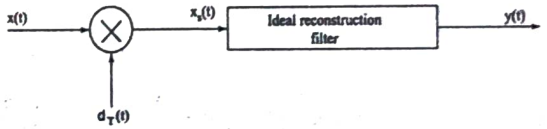


Fig. 9.25

The output of the ideal reconstruction filter can be found by sending signal $x_s(t)$ through an ideal lowpass filter having characteristics shown in Fig. 9.25.

For the following signals below draw the spectrum of $|X_s(j\Omega)|$ and find expression for $x(nT)$ and $y(t)$

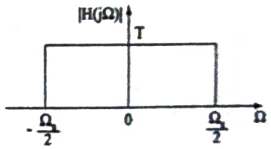


Fig. 9.25

(i) $x(t) = 2 + \cos(100\pi t)$ for $T = 0.0125$

(ii) $x(t) = 2 + \cos(100\pi t)$ for $T = \frac{1}{150}$ sec

(iii) $x(t) = 1 + \cos(10\pi t) + \cos(30\pi t)$ for $T = 0.04$ sec

(iv) If $X(j\Omega) = \frac{1}{2 + j\Omega}$ and $T = 2$ sec. Draw $|X_s(j\Omega)|$. Test for aliasing.

Solution:

Given $x(t) = 2 + \cos(100\pi t)$

$$= 2 + \frac{1}{2}\left[e^{j100\pi t} + e^{-j100\pi t}\right] \qquad \boxed{F[1] = 2\pi\delta(\Omega)}$$

$$X(j\Omega) = 4\pi\delta(\Omega) + \pi[\delta(\Omega + 100\pi) + \delta(\Omega - 100\pi)]$$

$$\boxed{F\left[e^{j\Omega_0 t}\right] = 2\pi\delta(\Omega + \Omega_0)}$$

The amplitude spectrum $|X(j\Omega)|$ is shown in Fig 9.26.

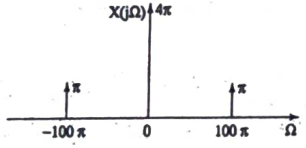


Fig. 9.26

The sampling time $T = 0.0125$

$$f_s = \frac{1}{T} = 80\,\text{Hz}$$

$$\Omega_s = 2\pi f_s = 160\,\text{rad/sec}$$

Given $f_m = 50\,\text{Hz}$

Nyquist rate $= 100\,\text{Hz}$

The sampling frequency is less than Nyquist rate. Therefore the successive spectra in $X_s(j\Omega)$ overlap and the signal $x(t)$ cannot be recovered from its samples.

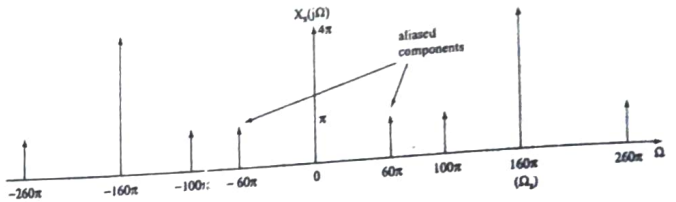


Fig. 9.27

The spectra $X_s(j\Omega)$ of the sampled signal $x(t)$, is periodic repetition of $X(j\Omega)$ for every 160π rad/sec (80 Hz). The spectra $X_s(j\Omega)$ is as shown in Fig. 9.27. The frequency characteristics of the reconstruction filter is shown in Fig. 9.28. When the signal with spectrum $|X_s(j\Omega)|$ shown in Fig. 9.27 is passed through the reconstruction filter (lowpass filter) the output of the filter consists of the frequency components as shown in Fig. 9.29.

That is

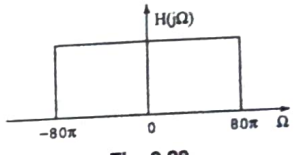


Fig. 9.28

$$Y(j\Omega) = 4\pi\delta(\Omega) + \pi\left[\delta(\Omega + 60\pi) + \delta(\Omega - 60\pi)\right]$$

$$\Rightarrow \quad y(t) = 2 + \cos(60\pi t)$$

$$x(nT) = x(t)\big|_{t=nT} = 2 + \cos(100\pi t)\big|_{t=0.0125\,n}$$

$$= 2 + \cos(1.25\,n\pi)$$

$$= 2 + \cos(0.75\,n\pi)$$

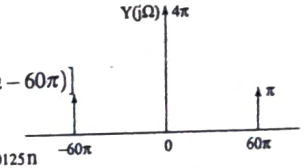


Fig. 9.29

ii) $x(t) = 2 + \cos(100\pi t)$ $T = \frac{1}{150}$ sec;

$$x(nT) = x(t)\big|_{t=nT}$$

$$x(nT) = 2 + \cos(100\pi t)\big|_{t=1/150} = 2 + \cos(0.666\,n\pi)$$

$$f_m = 50\,\text{Hz} \qquad f_s = 150\,\text{Hz}; \qquad \Omega_s = 2\pi f_s = 300\pi$$

The amplitude spectrum is shown in Fig. 9.26.

The spectrum of sampled signal of $x(t)$ is periodic extention of $X(j\Omega)$ for every 300π rad/sec (150 Hz) as shown in Fig. 9.30.

The spectrum $X_s(j\Omega)$ is free from aliasing, as the sampling rate $f_s(150Hz > 100Hz)$.

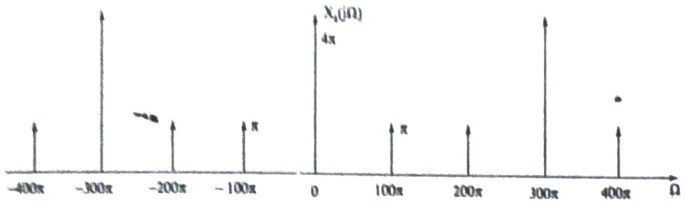


Fig. 9.30

The frequency characteristics of reconstruction filter is shown in Fig. 9.31. If the signal $x(nT)$ with spectrum $X_s(j\Omega)$ is passed through the reconstruction filter then the resultant output spectrum $Y(j\Omega)$ is as shown in Fig. 9.32.

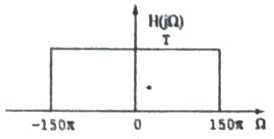


Fig. 9.31

$$Y(j\Omega) = 4\pi\delta(\Omega) + \pi\left[\delta(\Omega - 100\pi) + \delta(\Omega + 100\pi)\right]$$

$$\Rightarrow \quad y(t) = 2 + \cos(100\pi t)$$

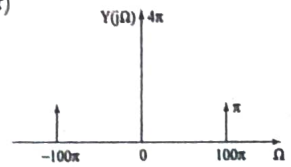


Fig. 9.32

iii) $x(t) = 1 + \cos(10\pi t) + \cos(30\pi t)$

$T = 0.04$ sec

$$x(nT) = 1 + \cos(0.4n\pi) + \cos(1.2n\pi)$$

$$f_s = \frac{1}{T} = 25\,\text{Hz}, \qquad \Omega_s = 50\pi\,\text{rad/sec}$$

$$\Omega_m = 2\pi f_m = 30\pi \Rightarrow f_m = 15\,\text{Hz}$$

Nyquist rate = 30 Hz

The sampling frequency $f_s < 2f_m$. Therefore the successive spectra of $X(j\Omega)$ in $X_s(j\Omega)$ overlaps and $x(t)$ cannot be recovered from $x(nT)$.

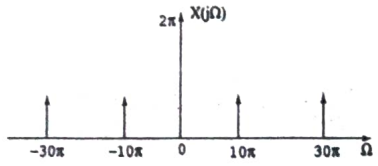


Fig. 9.33

$$X(j\Omega) = 2\pi\delta(\Omega) + \pi\left[\delta(\Omega + 10\pi) + \delta(\Omega - 10\pi)\right] + \pi\left[\delta(\Omega + 30\pi) + \delta(\Omega - 30\pi)\right]$$

$\Omega_s = 50\pi$ rad/sec. Therefore the spectrum of $X_s(j\Omega)$ can be obtained by periodically repeating $X(j\Omega)$ for every 50π rad/sec (25 Hz). The spectrum $X_s(j\Omega)$ is shown in Fig. 9.34.

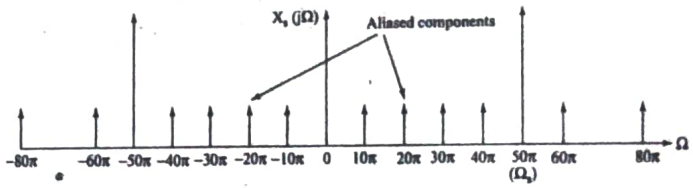


Fig. 9.34

The frequency spectrum of the lowpass filter is shown in Fig. 9.29.

The output spectrum of lowpass filter is shown in Fig. 9.36.

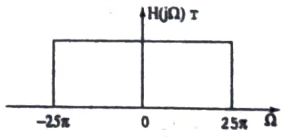


Fig. 9.35

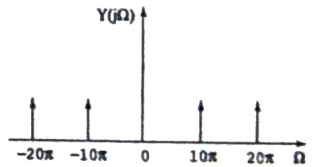


Fig. 9.36

$$\Rightarrow \quad y(t) = 1 + \cos(10\pi t) + \cos(20\pi t)$$

iv) Given $T = 2\,\text{sec}$ $f_s = 0.5Hz$

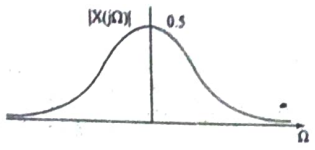


Fig. 9.37

$\Omega_s = \pi\,\text{rad/sec}$

$$X(j\Omega) = \frac{1}{2+j\Omega}$$

$$|X(j\Omega)| = \frac{1}{\sqrt{4+\Omega^2}}$$

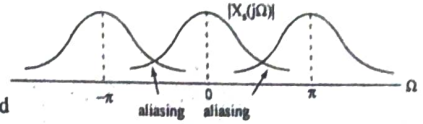


Fig. 9.38

The spectrum $|X(j\Omega)|$ is plotted in Fig. 9.37.

The spectrum $|X(j\Omega)|$ is not band limited. Therefore aliasing of successive frequency components occurs even Ω_s is large. The spectra $|X_s(j\Omega)|$ is shown in Fig. 9.38.

Practice Problem 9.2

Consider a continuous time signal $x(t)$ with frequency spectrum shown in Fig. 9.39. Find the frequency spectra of its sampled sequences if the sampling frequency

(i) $\Omega_s = 20\,\text{rad/sec}$
(ii) $\Omega_s = 40\,\text{rad/sec}$
(iii) $\Omega_s = 60\,\text{rad/sec}$

Also find in which cases the signal $x(t)$ can be recovered from its samples.

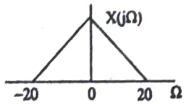


Fig. 9.39

Solved Problem 9.8 Determine the Nyquist sampling rate and Nyquist sampling intervals for the following signals

(i) $\text{sinc}^2(200\pi t)$ AU Nov 2009
(ii) $0.5\,\text{sinc}^2(200\pi t)$
(iii) $\text{sinc}(200\pi t) + 3\,\text{sinc}^2(120\pi t)$
(iv) $\text{sinc}(100\pi t)\,\text{sinc}(200\pi t)$

Solution:

We know

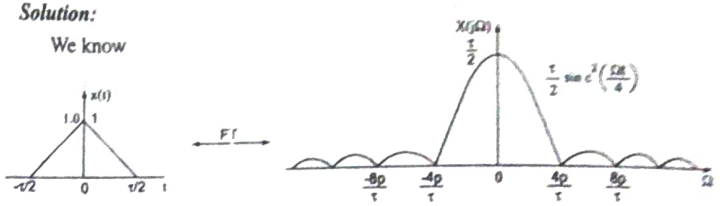


Fig. 9.40

Using Dual property, we have

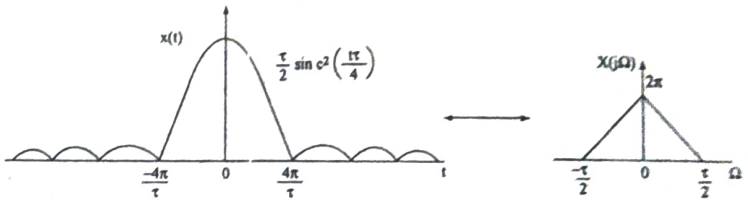


Fig. 9.41

Simflarly Fourier transform of a rectangular pulse is a *sinc* function

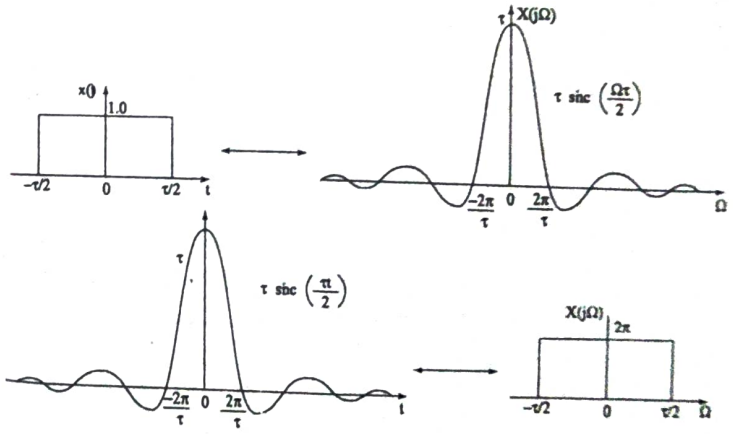


Fig. 9.42

(i) Given

$$x(t) = \text{sinc}^2(200\pi t)$$

Comparing the signal $x(t)$ with the signal in Fig. 9.35 we can find

$$\frac{t\tau}{4} = 200\pi t$$

$$\Rightarrow \tau = 800\pi$$

$$\tau/2 = 400\pi$$

That is the maximum frequency component present in the signal is 400π rad/sec (or) the signal is band limited to 400π rad/sec

$$\Omega_m = 400\pi$$

$$f_m = 200\,\text{Hz}$$

$$\text{Nyquist rate } f_s = 2f_m = 400\,\text{Hz}$$

$$\text{Sampling interval } T = \frac{1}{400} = 2.5\,\text{m sec}$$

(ii) Same value as above

(iii) Given $x(t) = \text{sinc}(200\pi t) + 3\,\text{sinc}^2(120\pi t)$

Let $x_1(t) = \text{sinc}(200\pi t)$ and $x_2(t) = 3\,\text{sinc}^2(120\pi t)$

Comparing the signal $x_1(t)$ with the signal in Fig. 9.34, we get

$$\frac{\tau}{2} = 200\pi \text{ rad/sec}$$

That is the signal $x_1(t)$ is band limited to $\Omega_{m_1} = 200\pi$ rad/sec

$$\Rightarrow \quad f_{m1} = 100\,\text{Hz}$$

For the signal $x_2(t)$

$$\tau/4 = 120\pi$$

$$\Rightarrow \tau/2 = 240\pi$$

The signal $x_2(t)$ is band limited to $\Omega_{m2} = 240\pi$ rad/sec.

When both signals are added the maximum frequency component

$$\Omega_m = \Omega_{m2} = 240\pi \text{ rad/sec}$$

$$f_m = 120\,\text{Hz}$$

$$\text{Nyquist rate } f_s = 2f_m = 240\,\text{Hz}$$

$$\text{Nyquist interval } T_s = \frac{1}{240} = 4.167\,\text{m sec}$$

(iv) $x(t) = \text{sinc}(100\pi t)\,\text{sinc}(200\pi t)$

Let $x_1(t) = \text{sinc}(100\pi t); x_2(t) = \text{sinc}(200\pi t)$

The signal $x_1(t)$ is band limited to frequency 100π rad/sec and the signal $x_2(t)$ is band limited to frequency 200π rad/sec.

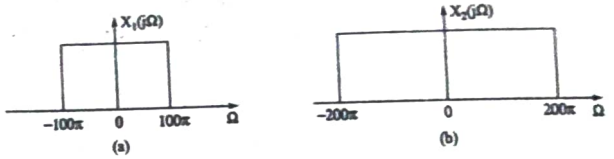


Fig. 9.43

The frequency spectra of $x(t)$ can be obtained by convolving the individual spectra of $x_1(t)$ and $x_2(t)$. That is

$$F[\text{sinc}(100\pi t)\,\text{sinc}(200\pi t)] = X_1(j\Omega) * X_2(j\Omega)$$

The convolution of $X_1(j\Omega)$ and $X_2(j\Omega)$ results a triangular shape frequency spectra with $\Omega_m = 300\pi$ rad/sec as shown in Fig. 9.44.

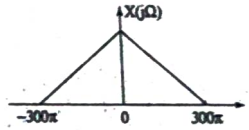


Fig. 9.44

Therefore

$$f_m = 150\,\text{Hz}$$

$$\text{Nyquist rate} = 2f_m = 300\,\text{Hz}$$

$$\text{Nyquist interval} = \frac{1}{300} = 3.33\,\text{m sec}$$

Practice Problem 9.3

1. The amplitude spectrum of a continuous time signal is shown in Fig. 9.45. If the signal in figure is sampled with a sampling frequency of $\Omega_s = 200\pi$ rad/sec, sketch the amplitude spectrum of the sampled signal

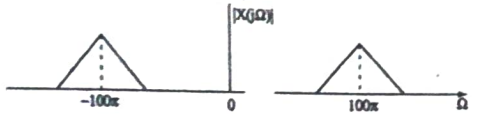


Fig. 9.45

Practice Problem 9.4

1. The amplitude spectrum of $x(t)$ is shown in Fig. 9.46. This signal is amplitude modulated by $\cos(75\pi t)$. If the modulated signal is sampled with sampling period $T = 0.01$ sec, sketch the amplitude spectrum of the sampled signal.

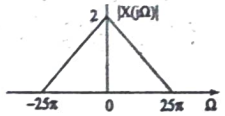


Fig. 9.46

9.10 Bandpass Signals

A bandpass signal $x(t)$ is a signal whose frequency response $|X(j\Omega)| = 0$ for $|\Omega - \Omega_0| \geq W$ where $W < \Omega_0$. The frequency Ω_0 is usually referred to as the central frequency of the bandpass signals. In communication system the modulated signals are examples for bandpass signals. One such waveform is shown in Fig.9.47.

9.11 Sampling Bandpass Signals

For a signal $x(t)$ whose highest frequency component is f_m and low frequency component $f_L = 0$, the sampling frequency must be greater than equal to $2f_m$. In general for the case, where the low frequency component $f_L \neq 0$, there will be no aliasing if $f_s > 2f_H$. Also there might be no aliasing even if $f_s = 2(f_H - f_L)$. To prove the above condition let us consider a bandpass signal whose spectrum is

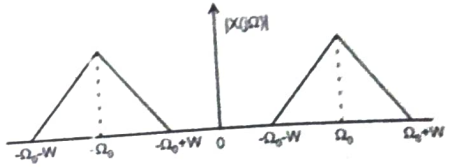


Fig. 9.47

shown in Fig.9.48. Let f_H and f_L are the highest and lower frequency components present in the signal. Let us select the sampling frequency $f_s = 2[f_H - f_L]$. For our convenience let the low frequency component f_L is integral multiple of f_s. That is $f_L = nf_s$. If the sampling frequency is exactly $f_s = 2[f_H - f_L]$ the resultant spectrum is shown in Fig.9.49. The spectra in Fig.9.49 is derived from the spectra of $x(t)$ shown in Fig.9.48 by shifting the pattern in to the right and also to the left by an amount f_s. Similarly the highest harmonics of signal $x(t)$ give rise to corresponding shift, right and left, of the spectra $X(j\Omega)$. From the spectra we can observe that by passing the sampled signal through a bandpass filter with arbitrary sharp cut of f_s and with pass band from f_L to f_H, the signal $x(t)$ will be recovered.

The spectrum of $x(t)$ extends between $2f_s$ and $2.5f_s$ therefore, there is no spectral overlap. Similarly if the spectrum of $x(t)$ extends between $2.5f_s$ to $3f_s$, there will be no spectral overlap. However if the spectrum of $X(j\Omega)$ were not confined either to first half or to the second-half of the interval between sampling frequency, there would be overlap between the spectra pattern and signal recovery would not be possible. Hence the minimum sampling frequency allowable is $f_s = 2[f_H - f_L]$ provided that either f_H or f_L is a harmonic of f_s.

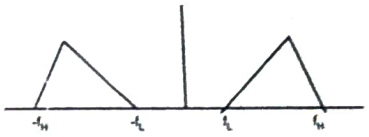


Fig.9.48

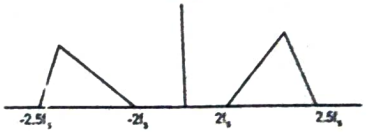


Fig. 9.49

Practice Problem 9.3

1. The amplitude spectrum of a continuous time signal is shown in Fig. 9.45. If the signal in figure is sampled with a sampling frequency of $\Omega_s = 200\pi$ rad/sec, sketch the amplitude spectrum of the sampled signal

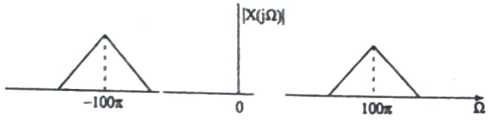


Fig. 9.45

Practice Problem 9.4

1. The amplitude spectrum of $x(t)$ is shown in Fig. 9.46. This signal is amplitude modulated by $\cos(75\pi t)$. If the modulated signal is sampled with sampling period $T = 0.01$ sec, sketch the amplitude spectrum of the sampled signal.

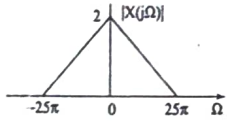


Fig. 9.46

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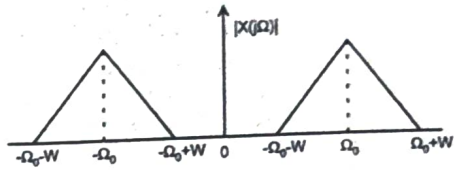


Fig. 9.47

shown in Fig.9.48. Let f_H and f_L are the highest and lower frequency components present in the signal. Let us select the sampling frequency $f_s = 2[f_H - f_L]$. For our convenience let the low frequency component f_L is integral multiple of f_s. That is $f_L = nf_s$. If the sampling frequency is exactly $f_s = 2[f_H - f_L]$ the resultant spectrum is shown in Fig.9.49. The spectra in Fig.9.49 is derived from the spectra of $x(t)$ shown in Fig.9.48 by shifting the pattern in to the right and also to the left by an amount f_s. Similarly the highest harmonics of signal $x(t)$ give rise to corresponding shift, right and left, of the spectra $X(j\Omega)$. From the spectra we can observe that by passing the sampled signal through a bandpass filter with arbitrary sharp cut of f_s and with pass band from f_L to f_H, the signal $x(t)$ will be recovered.

The spectrum of $x(t)$ extends between $2f_s$ and $2.5f_s$ therefore, there is no spectral overlap. Similarly if the spectrum of $x(t)$ extends between $2.5f_s$ to $3f_s$, there will be no spectral overlap. However if the spectrum of $X(j\Omega)$ were not confined either to first half or to the second-half of the interval between sampling frequency, there would be overlap between the spectra pattern and signal recovery would not be possible. Hence the minimum sampling frequency allowable is $f_s = 2[f_H - f_L]$ provided that either f_H or f_L is a harmonic of f_s.

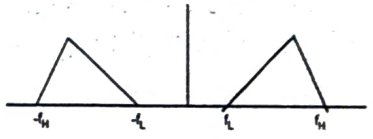


Fig.9.48

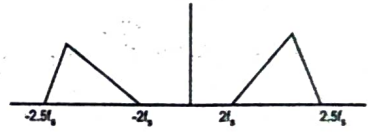


Fig. 9.49

If neither f_H or f_L is a harmonic of f_s, a more general analysis is required. The spectrum of $x(t)$ is redrawn as shown in Fig.9.50. Now let us find the constraint that must be imposed so that there is no overlap of spectra. Let the positive frequency and negative frequency part of the spectrum are called PS and NS respectively. When we sample the signal the NS and PS pattern will shift right or left and overlap depending on sampling rate. If we select the minimum value of $f_s = 2(f_H - f_L)$ then shifted PS patterns will not overlap with PS. The shifted pattern of *NS* to the left of NS will not overlap with PS. However the right shifted pattern of NS cause an overlap.

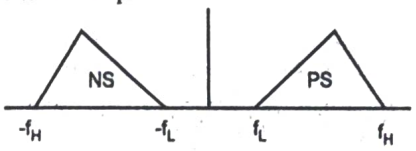


Fig. 9.50

Fig.9.51 shows the right-shift pattern of NS due to the $(N-1)$st and N^{th} harmonic of the sampling wave form

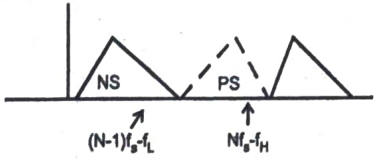


Fig. 9.51

To avoid overlap

$$(N-1)f_s - f_L \le f_L$$

$$\Rightarrow Nf_s - f_H > f_L$$

$$\text{Let } B = f_H - f_L$$

$$(N-1)f_s \le 2(f_H - B)$$

$$\text{and } Nf_s \ge 2f_H$$

$$\text{Let } K = \frac{f_H}{B}, \text{ then } f_s \le 2B\left(\frac{k-1}{N-1}\right)$$

$$\text{and } f_s \ge 2B\left(\frac{k}{N}\right)$$

in which $k \ge N$ since $f_s \ge 2B$

From the above discussion we can state sampling theorem for band pass signal as below.

A band pass signal with highest frequency f_H and bandwidth B, can be recovered from its samples through bandpass filtering by sampling it with frequency $f_s = \frac{2f_H}{k}$ where k is the largest integer not exceeding f_H/B.

Short Questions and Answers

1. **State sampling theorem.**

   A band limited signal $x(t)$ with $X(j\Omega) = 0$ for $|\Omega| > \Omega_m$ is uniquely determined from its samples $x(nT)$, if the sampling frequency $f_s \ge 2f_m$, that is, sampling frequency must be atleast twice the highest frequency present in the signal.

2. **Define Nyquist rate.**

   The frequency $2f_m$, which under sampling theorem must be exceeded by the sampling frequency is known as the Nyquist rate.

3. **What is an anti-aliasing filter?**

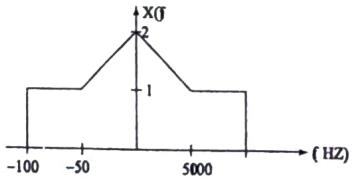
   A filter that is used to reject high frequency signals before it is sampled to reduce the aliasing is called an antialiasing filter.

4. **What is the transfer function of a zero order hold?**

$$\frac{1 - e^{-sT}}{s}$$

5. **A signal $x(t)$, whose spectrum is shown in Fig. Q9.1 is sampled at a rate of 300 samples/sec. What is the spectrum of the sampled discrete-time signal?**

   AU April 2003



Ans

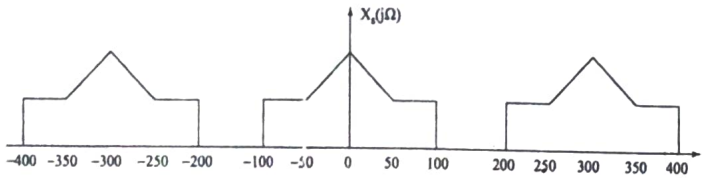
$$f_m = 100\,Hz$$

$$\text{Nyquist rate} = 2f_m = 200\,\text{Hz}$$

$$\text{Sampling frequency} = f_s = 300\,\text{Hz}$$

$f_s > 2f_m$. Therefore no aliasing.

The spectrum of the sampled signal repeats for every 300 Hz.



Multiple Choice Questions

1. Flat-top sampling of lowpass signals
   (a) give rise to aperture effect (b) implies oversampling
   (c) leads to aliasing (d) Introduces delay distortion

2. The Nyquist sampling frequency (in Hz) of a signal given by $6 \times 10^4 \sin c(400\pi t$ $10^6 \sin c(100\pi t)$ is
   (a) 200 (b) 300 (c) 500 (d) 1000

3. A signal $x(t) = 100\cos(24\pi \times 10^3(t)$ is ideally sampled with a sampling period of 50μ sec and then passed through an ideal lowpass filter with cutoff frequency of 15 kHz. Which of the following frequencies are present at the filter output?
   (a) 12 kHz only (b) 8 kHz only (c) 12 kHz and 9 kHz (d) 12 kHz and 8 kHz

4. The Nyquist sampling rate for the signal $g(t) = 10\cos(50\pi t)\cos^2(150\pi t)$ where t is in seconds is
   (a) 150 samples/sec (b) 200 samples/sec (c) 300 samples/sec (d) 350 samples/sec

5. A signal $x(t) = 6\cos 10\pi t$ is sampled at the rate of 14 Hz. To recover the original signal, the cutoff frequency f_c of the ideal low-pass filter should be
   (a) $5\text{Hz} < f_c < 9\text{Hz}$ (b) 9Hz (c) 10Hz (d) 14Hz

6. What is Nyquist sampling rate of the following signal
   $$x(t) = \sin c^2(100\pi t)$$
   (a) 50 Hz (b) 100 Hz (c) 200 Hz (d) 400 Hz

Answers to Multiple Choice Questions

1. (a)
2. (c)
3. (a)
4. (d)
5. (a)
6. (c)